

Program Evaluation with Unconfounding Treatment

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Unconfounding Treatment

- Assume that treatment is **randomly assigned conditional on observable covariates**.
 - “ignorability of treatment”, “selection on observables”, “exogenous treatment”.
- Let $T_i \in \{0, 1\}$ denote treatment status, and let $Y_i(1), Y_i(0)$ be the potential outcomes.
- The *propensity score* is the probability of treatment given observables:

$$p(X_i) \equiv \Pr(T_i = 1 \mid X_i) = \Lambda(X_i' \beta),$$

where $\Lambda(\cdot)$ is typically the logit or probit link function.

- **Strong ignorability**

$$(Y_i(1), Y_i(0)) \perp T_i \mid X_i, \quad 0 < p(X_i) < 1$$

Note: conditioning directly on X_i is difficult if X_i is high-dimensional.

- **Rosenbaum & Rubin 1983:** If selection on observables holds conditional on X_i , it holds conditional on $p(X_i)$.

$$(Y_{i1}, Y_{i0}) \perp T_i \mid p(X_i)$$

Nearest-Neighbor Matching on the Propensity Score

- First estimate the propensity score, obtaining

$$\hat{p}_i = \Lambda(X_i' \hat{\beta}).$$

- For each treated individual i with $T_i = 1$, choose an untreated match $j(i)$ with $T_{j(i)} = 0$ such that

$$j(i) \in \arg \min_{j: T_j=0} |\hat{p}_j - \hat{p}_i|.$$

- The matched untreated outcome proxies for the missing counterfactual:

$$Y_i(0) \approx Y_{j(i)}.$$

- Hence the estimated treatment effect for treated unit i is

$$\hat{\tau}_i^{ATT} = Y_i - Y_{j(i)}.$$

Treatment Effects from Matching

- The **average treatment effect on the treated (ATT)** is estimated by

$$\widehat{ATT} = \frac{1}{N_1} \sum_{i: T_i=1} (Y_i - Y_{j(i)}),$$

where $N_1 = \sum_{i=1}^n T_i$ is the number of treated units.

- Similarly, the **average treatment effect on the untreated (ATU)** is obtained by, for each untreated individual, finding the closest treated match.
- A sample analogue of the **ATE** can then be formed by averaging matched differences over treated and untreated units.

Dehejia & Wahba (1999)

- **Context:** Lalonde (1986) evaluated whether nonexperimental econometric methods could replicate results from a randomized experiment.
- **The Finding:** Standard nonexperimental estimators (regression, fixed-effects, etc.) were inaccurate or highly sensitive to specification.
- **The Question:** Can we do better? Dehejia & Wahba (1999) revisit this data to see if propensity score methods can bridge the gap.
- **Key Insight:** Propensity scores offer a powerful diagnostic for comparability between treated and nonexperimental comparison units.

Data and Empirical Setting

- **Treatment (T):** National Supported Work (NSW) Demonstration, a labor training program for *individuals facing economic/social problems*.
- **Target Estimand Outcome (Y):** Post-intervention earnings in 1978 ($RE78$).
- **The Data Structure:**
 - *Treatment Group:* Male NSW participants (experimental).
 - *Control Group:* NSW randomized controls (provides the benchmark).
 - *Comparison Groups:* Nonexperimental survey datasets from PSID and CPS.
- **The Pre-intervention Variables (X):** Age, education, ethnicity, marital status, and pre-intervention earnings ($RE74$, $RE75$).

Why Naïve Comparisons Fail

- **Massive Baseline Differences:** PSID and CPS comparison groups differ dramatically from the NSW treated group in age, earnings, and demographics.
- **Naive Estimates:** A simple difference in means yields heavily negative treatment effects for most comparison groups (e.g., $-\$15,205$ for PSID-1).
- **Regression Failure:** OLS models controlling for observables remain sensitive and often far from the experimental benchmark.
- **The Core Issue:** Linear regression extrapolates across areas where the treatment and comparison groups have no common support.

Balance Checks

- **Step 1: Estimate $p(X_i)$.** The paper uses a logistic probability model, estimated separately for each comparison sample.
- **Step 2: Balance Property.** The propensity score is a *balancing score*:

$$X_i \perp T_i \mid p(X_i)$$

- **Checking Balance:** Group observations into strata defined by $p(X_i)$. Test if first and second moments of X_i are equal across $T = 1$ and $T = 0$ within each stratum.
- **Iteration:** If balance is rejected, add higher-order terms or interactions to the logit model and try again.

Common Support / Overlap Assumption (Figure 1 & Figure 2)

- **The Overlap Assumption:** Every unit must have a non-zero, non-one probability of receiving treatment.

$$0 < p(X_i) < 1$$

- **Diagnostic Value:** Comparing histograms of $p(X_i)$ for treated vs. comparison groups quickly reveals the extent of overlap.
- **Trimming:** The paper explicitly discards comparison units whose estimated $p(X_i)$ falls below the minimum (or above the maximum) of the treated units.
- **Paper finding:** Discarding thousands of non-overlapping comparison units dramatically improves comparability.
- Ad-hoc selection of comparison groups is inferior to systematically matching on all variables simultaneously via $p(X_i) \Rightarrow$ Highly efficient dimension reduction!.

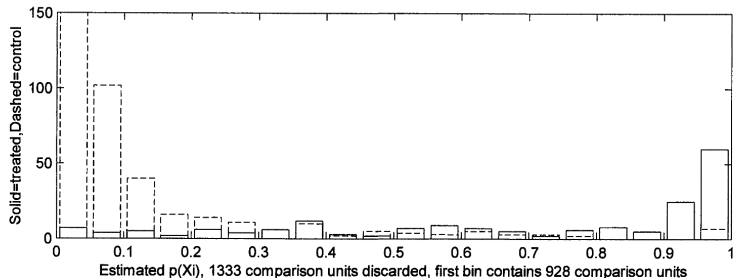


Figure 1. Histogram of the Estimated Propensity Score for NSW Treated Units and PSID Comparison Units. The 1,333 PSID units whose estimated propensity score is less than the minimum estimated propensity score for the treatment group are discarded. The first bin contains 928 PSID units. There is minimal overlap between the two groups. Three bins (.8-.85, .85-.9, and .9-.95) contain no comparison units. There are 97 treated units with an estimated propensity score greater than .8 and only 7 comparison units.

Main Empirical Results (Table 3)

- **Experimental Benchmark Estimate:** \$1,794.
- **Naïve Difference in Means:** $-\$15,205$ for PSID-1 and $-\$8,498$ for CPS-1.
- **Propensity Score Stratification:**
 - PSID-1: \$1,608.
 - CPS-1: \$1,713.
- **Propensity Score Matching:**
 - PSID-1: \$1,691.
 - CPS-1: \$1,582.
- **Conclusion:** Propensity score estimates are remarkably close to the unbiased experimental benchmark.

Sensitivity Analysis & Limitations (Table 5)

- **Sensitivity to Score Specification:** The estimates are relatively robust to different logit specifications, provided the specification achieves covariate balance.
- **Sensitivity to Observables:** The estimators are highly sensitive to the *selection on observables* assumption.
 - **The Importance of Pre-treatment History:** Dropping 1974 earnings (*RE74*) drastically degrades the estimates, proving that rich pre-treatment histories are required for ignorability to hold.
 - **Use Multiple Comparison Groups.** If your ATT estimate varies wildly between two different valid comparison groups (e.g., PSID vs CPS), you likely have an unobservables problem.
- **Limitation:** Propensity scores only balance observables. They cannot fix selection on unobservables. (Still a useful benchmark though.)

References

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Imbens, Guido W. “Causal inference in the social sciences.” *Annual Review of Statistics and Its Application* 11 (2024).

Rosenbaum, Paul R., and Donald B. Rubin. “The central role of the propensity score in observational studies for causal effects.” *Biometrika* 70.1 (1983): 41-55.

Exercise - IV

Consider the structural equation

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + e \quad (1)$$

with $X \in \mathbb{R}$ treated as endogenous so that $\mathbb{E}[Xe] \neq 0$. We have an instrument $Z \in \mathbb{R}$ which satisfies $\mathbb{E}[e | Z] = 0$ so in particular $\mathbb{E}[e] = 0$, $\mathbb{E}[Ze] = 0$ and $\mathbb{E}[Z^2 e] = 0$.

- ① Should X^2 be treated as endogenous or exogenous?
- ② Suppose

$$X = \gamma_0 + \gamma_1 Z + u \quad (2)$$

with u independent of Z and mean zero. Consider using $(1, Z, Z^2)$ as instruments. Is the problem just-identified, over-identified, or under-identified?

- ③ Write out the reduced form equation for X^2 . Under what condition on the reduced form parameters in (2) are the parameters in (1) identified?

Solution

1. Yes.
2. There are 2 endogenous regressors (X and X^2) and 2 excluded instruments (Z and Z^2), so the model is just-identified.

Solution

$$\begin{aligned} X^2 &= (\gamma_0 + \gamma_1 Z + u)^2 \\ &= \gamma_0^2 + 2\gamma_0\gamma_1 Z + \gamma_1^2 Z^2 + 2\gamma_0 u + 2\gamma_1 Z u + u^2. \\ &= \pi_0 + \pi_1 Z + \pi_2 Z^2 + v, \end{aligned}$$

where

$$\pi_0 = \gamma_0^2 + \mathbb{E}[u^2], \quad \pi_1 = 2\gamma_0\gamma_1, \quad \pi_2 = \gamma_1^2,$$

and

$$v = 2\gamma_0 u + 2\gamma_1 Z u + (u^2 - \mathbb{E}[u^2]).$$

Because $u \perp Z$ and $\mathbb{E}[u] = 0$,

$$\mathbb{E}[v \mid Z] = 0.$$

Solution

Assuming Z^2 is not collinear with 1 and Z .

counterexample: Z is a dummy variable $\Rightarrow Z = Z^2$, or Z is a constant.

$$\begin{pmatrix} 1 \\ X \\ X^2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ \gamma_0 & \gamma_1 & 0 \\ \gamma_0^2 + \mathbb{E}[u^2] & 2\gamma_0\gamma_1 & \gamma_1^2 \end{pmatrix}}_{\Pi} \begin{pmatrix} 1 \\ Z \\ Z^2 \end{pmatrix} + \begin{pmatrix} 0 \\ u \\ 2\gamma_0 u + 2\gamma_1 Z u + (u^2 - \mathbb{E}[u^2]) \end{pmatrix} \cdot,$$

$$\det(\Pi) \neq 0 \iff \gamma_1 \neq 0$$

Exercise - Probit

Suppose that $(X_i, \varepsilon_i)'$ i.i.d jointly normal with mean zero, covariance matrix $\begin{pmatrix} 1 & 0 \\ 0 & \sigma^2 \end{pmatrix}$. Suppose moreover that $Y_i^* = \beta X_i + \varepsilon_i$. Define the binary variable I_i as 1 if $Y_i^* > 0$ and 0 otherwise.

- 1 Show how to estimate β/σ by a probit regression in which I_i is the dependent variable.

Solution

Because (X_i, ε_i) are jointly normal and uncorrelated, they are independent:

$$\varepsilon_i \perp X_i.$$

Hence, conditional on $X_i = x$,

$$Y_i^* | X_i = x \sim N(\beta x, \sigma^2).$$

Therefore,

$$\begin{aligned}\Pr(I_i = 1 | X_i = x) &= \Pr(Y_i^* > 0 | X_i = x) \\ &= \Pr(\beta x + \varepsilon_i > 0 | X_i = x) \\ &= \Pr\left(\frac{\varepsilon_i}{\sigma} > -\frac{\beta}{\sigma}x\right) = \Phi\left(\frac{\beta}{\sigma}x\right)\end{aligned}$$

Thus β/σ is estimated by running a probit regression of I_i on X_i *without an intercept*. The probit coefficient is exactly β/σ .